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11 May 1988
SUPERSEDING
MIL-P-52910A(ME)
8 March 1979

MILITARY SPECIFICATION

PALLET: CARGO

FOR RIBBON BRIDGE TRANSPORTER

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cargo pallets capable of functioning with the ribbon bridge transporter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to be extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-Marking for Shipment and Storage.
MIL-STD-209	-Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-889	-Dissimilar Metals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13221E6586 - Pallet: Cargo, For Ribbon Bridge, Transporter.
13226E7234 - Camouflage Pattern and Inspection.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY

A5.10 - Aluminum and A1uminum Alloy Welding Rods and Bare Electrodes.
D1.2 - Structural Welding Code - Aluminum

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The pallet shall be in accordance with Top Assembly TA13221E6586 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the pallet. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and on the applicable drawings. Any material not specified shall be selected by the contractor and shall be subject to all provisions on this specification.

3.3.1 Material deterioration prevention and control. The pallets shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the pallets may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the pallets may be newly fabricated from recovered materials to the maximum extent practicable, provided the pallets produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the pallets.

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3.3.5 Welding electrodes for aluminum. Welding electrodes for aluminum shall conform to AWS A5.10, classification ER5356.

3.4 Slinging and tiedown provisions. Slinging and tiedown provisions shall be as specified on the drawings and conform to MIL-STD-209.

3.5 Identification marking. The pallet shall be identified as shown on the drawings.

3.6 Shipping data plate. A data plate shall be affixed to each pallet in accordance with the drawings.

3.7 Instruction plate. Each pallet shall be equipped with an instruction plate as specified on the drawings.

3.8 Treatment and painting. Treatment and painting shall be as specified on the drawings. Camouflage pattern painting shall be in accordance with drawing 13226E7234.

3.9 Lubrication. Lubrication shall be as shown on the end product drawings.

3.10 Government-loaned property. The following property shall be loaned by the Government (see 6.6):

a. Ribbon Bridge Transporter (MIL-T-52735) (transporter) - TM5-5420-209-12
b. Operation and Maintenance Manual (cargo pallet) - TM5-5420-208-128P

3.11 Workmanship.

3.11.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shal1 be done by methods that will not cause injury to the material. Corners shall be square and true. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. External surfaces shall be free from burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety. Aluminum shall not be cut by processes using heat.

3.11.2 Welders and welding.

3.11.2.1 Welders. Before assigning any welder or welding operator to a welding operation covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed the qualification test as specified by either the following listed codes for the type of welding operations to be performed effective as defined by the particular code:

ASME boiler and pressure vessel code, Section IX, Welding and Blazing Qualifications.

AWS D1.2 Structural Welding Code - Aluminum.

The contractor who makes only horizontal welds need not qualify welders or welding operators for "all position welding". The contractor is responsible for determining that automatic welding equipment operators are capable of

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producing quality welds in accordance with AWS and ASME codes. The test results shall be made available for review, upon request, by contracting officer or designated representative. In the event of evidence of poor welds, the Government reserves the right to require the retesting of any welder or welding operator.

3.11.2.2 Aluminum welding. Aluminum parts to be welded shal1 be sheared or sawed. Flame cutting shall not be used. Plasma-arc cutting is permissable if in conformance with AWS D1.2 section 3. The surfaces of aluminum parts to be welded shall be free from paint, grease, oxide film, and all other foreign matter, which shall be removed using a stainless steel wire brush. The welding shall be done by Gas Metal Arc Welding (GMAW) or Gas Tungsten Arc Welding (GTAW) method. Welds may be ground, filed, buffed, or chipped. Welds shall not be hammered. Completed welds shall be free from discontinuity such as base metal cracks adjacent or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal, lack of fusion with parent metal of between weld passes, lack of penetration, lack of prescribed fit, or splatter or code requirements.

3.12 Durability. The pallet shall withstand, without failure, 100 launch and retrieve cycles as specified in 4.5.2.3. Failure criteria is defined in 4.5.2.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Quality Assurance Provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.7).

4.2 Classification of inspections.

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection procedure (see 4.5).
d. Inspection of packaging (see 4.6).

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4.3 First article inspection.

4.3.1 First article examination. The first article pallets shall be examined as specified in 4.5.1 in the order shown. The presence of one or more defects shall be cause for rejection.

4.3.2 First article test. The first article shall be tested as specified in 4.5.2.1, 4.5.2.2 and 4.5.2.3. Failure of any test shall be cause for providing corrective action to eliminate the deficiencies responsible for the failure. Continued failure of any test after corrective action has been taken shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Sample pallets shall be examined for the characteristics specified in 4.5.1.1. Presence of a defect shall be cause for rejection.

4.4.2 Tests. Sample pallets as selected in accordance with 4.5.1 shall be tested as specified in 4.5.2.1, 4.5.2.2 and 4.5.2.3. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Sampling. Sampling for examination and testing shall be in accordance with MIL-STD-105.

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4.5.1.1 Examination. Sample pallets selected in accordance with 4.5.1 shall be examined as specified herein for the following characteristics listed in table I. AQL shall be 1.0 percent for major defects.

TABLE I. Examination Schedule - Requirements.

NUMBER	CHARACTERISTICS	REQUIREMENT PARAGRAPH
MAJOR		
101.	Materials not as specified.	3.3
102.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
103.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
104.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.3
105.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the pallets.	3.3.4
106.	Welding electrodes for aluminum do not conform to AWS A5.10, classification ER5356.	3.3.5
107.	Slinging and tiedown provisions not as specified.	3.4
108.	Identification marking not as specified.	3.5
109.	Shipping data plate not as specified.	3.6
110.	Instruction plate not as specified.	3.7
111.	Lubrication not as specified.	3.9
112.	Metal fabrication not as specified.	3.11
113.	Welding not as specified.	3.11
114.	Machine work not as specified.	3.11
115.	Camouflage pattern data not as specified.	3.8
116.	Treatment and painting not as specified.	3.8

4.5.2 Tests.

4.5.2.1 Nesting. Nest two pallets making sure that the rear tiedown pins are locked under the nesting plates of the lower pallet and that the nesting pin can be inserted without forcing. Reverse the positions of the two pallets and nest the pallets once again, insuring that the nesting pin can be inserted freely.

4.5.2.2 Performance. Utilizing six or eight 40 x 48 inch general purpose pallets as shown in figure 1, place an evenly distributed 5-ton load on each pallet using cargo straps or ropes to secure cargo to pallet. Retrieve the loaded pallet with a ribbon bridge transporter in accordance with the Operation and Maintenance Manual provided with the transporter. Check the engagements of the pallet tiedown pin with the transporter rear hook and the forward locking pin. After successfully retrieving the loaded pallet return it to its original position by launching in accordance with the Operation and Maintenance Manuals (TM5-5420-209-12 and TM5-5420-208-128P).

4.5.2.3 Durability. Under the conditions shown on table II, launch and retrieve the pallet 100 times using a ribbon bridge transporter. For the portion of the test involving the 10 and 20 percent slopes, the transporter shall face the upward slope and the transporter and pallet shall be entirely on the slope during tests. For the 5-ton payload test, the loading arrangement shall be the same as selected for the performance test specified in 4.5.2.2.

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TABLE II. Durability test.

Payload	Transporter Position		
	20% Longitudinal Slope	10% Longitudinal Slope	Flat
Empty	5	5	10
5 Ton	5	5	35
Stacked (5 pallets)	5	5	25

4.5.2.4 Failure criteria. Evidence of permanent deformation, structural failure, weld failure, or inability of any part to properly perform its function shall constitute failure of the above tests. During the durability test only, repairs performed by organizational maintenance will be allowed except; failures showing a statistical pattern, even when repairable, shall constitute failure of the test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack. Each pack of pallets shall be inspected for the characteristics listed below. Presence of one or more defects shall be cause for rejection. Any defects shall be corrected prior to shipment.

117. Preservation and packing (see 6.9) not as specified, (see 5.1).
118. Pallets not bundled together to form a compact bundle, and secured properly, (see 5.1.1).
119. Marking (see 6.9) not as specified, (see 5.2).

5. PACKAGING

5.1 Preservation and packing (see 6.9).

5.1.1 Pallets. Three to seven pallets shall be bundled together to form a compact bundle. The bundle shall be secured by utilizing the incorporated securing devices.

5.2 Marking (see 6.9). Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cargo pallet is designed to mount on the ribbon bridge transporter to provide the transporter with cargo hauling capabilities.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of the first article (see 3.2).
c. When the Government will conduct any or all of the first article tests. When the Government will conduct a portion of the tests, the contracting officer should specify which test they will conduct (see 3.2).

6.3 First article. When a first article inspection is required, the items should be an initial production model. The first article should consist of two pallets. The contracting officer should include specific instructions

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in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manua1 to be furnished with each pallet.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of pallets.

6.6 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.10.

6.7 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirement, actual measurement, number and types of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable."

6.8 Definitions.

6.8.1 Quality assurance provisions {QAP). A QAP shown on the drawings is a contractual requirement that supplements section 4 of the specification. QAPs indicate the minimum requirements which must be inspected on the end product and assure interchangeability of repair parts.

6.9 Preservation and packing. Degrees of preservation and packing have not been included since the preservation specified is the minimum acceptable for any degree of protection and no packing is required. For purposes of preservation/packing level marking, the pallets shall be marked A/A.

6.10 Subject term(key word) listing.

Pallet
Pallet cargo
Pallet cargo, ribbon bridge transporter

6.11 Changes from previous issue. Asterisks (or vertica1 lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

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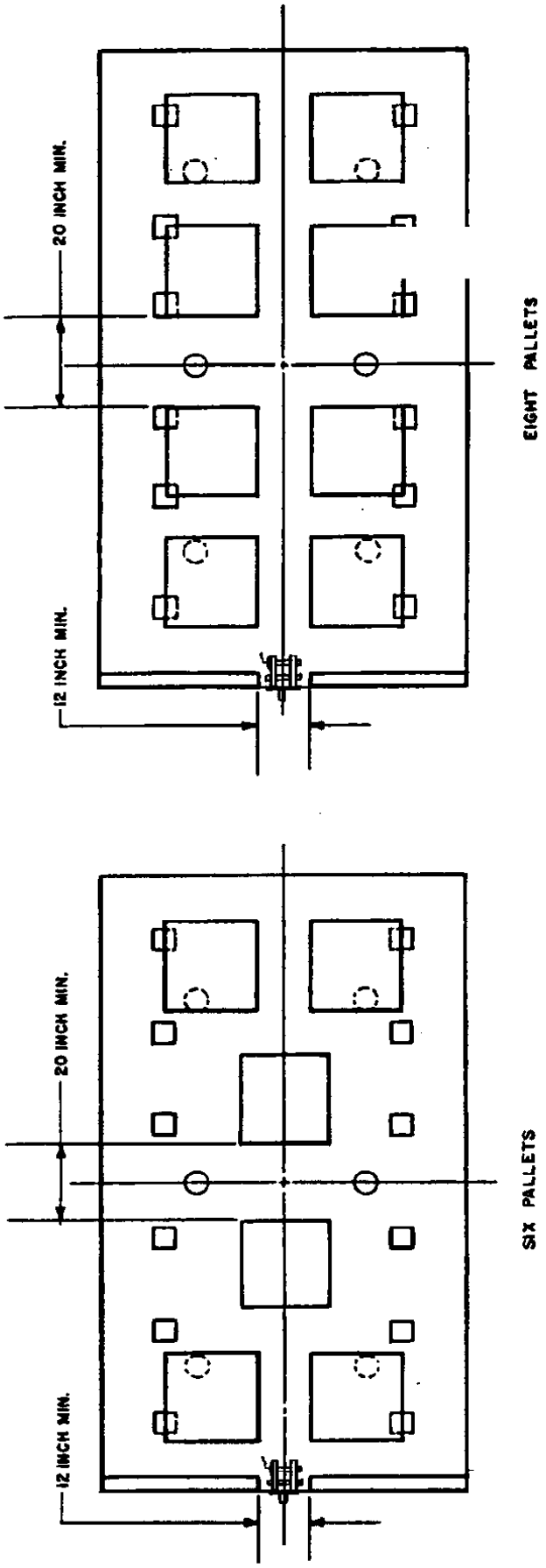

FIGURE 1. Loading arrangement of general purpose pallets.

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